



A Case Study on ICT Collaboration Performance in Automobile Business

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ABSTRACT

It is considered for companies to make up new types of organization, adopt ICT technologies tailored to the features of company, or etc, as a plan for the company to survive in the competitive environment. Especially, Utilizing IT technologies, when companies do their business, can make companies share their information, and it leads to reduce their costs in the business process and shorten their work time. Namely, introducing and utilizing ICT collaborative system that can make companies collaborate on their business brings about economic collaborative performance for the companies. So, a mother company and its partner companies bring ICT collaborative systems in and share their information that is needed by each company to secure their future value, and also try to lead the competitiveness of companies in the maximum level by constructing the collaborative network between a mother company and its partner companies.

In particular, the automobile industry has a great variety of steps such as materials, half-finished goods, work in process and product for producing the final product, and the number of the partner companies that supply half-finished goods is greater than other industries. In addition, type and number of its partner companies is diverse, and the parent company has a high level of reliance on its partner companies so that the need of the outsourcing collaborative system utilizing ICT is continuously increasing.

There have been studies on the level of ICT utilization, the level of ICT collaboration, and so on. However, few writers have been able to draw on any structured research into the study, performance analysis depending on ICT collaboration.

Therefore, in this study we tried to figure out the performance level measured from the ICT collaborative system constructed

based on the business processes such as product development, demand forecasting, production planning, distribution, and logistics management in the automobile industry, and to analyze which factors influence the level of performance.

In this study, we collected and analyzed basic data for the target of survey and business manager of the parent company but It is need to proceed analysis of business performance which is active and has validity from the step of business agreement to the step after the end of business rather than submit the data for performance tracking obligatorily for acquiring an objective performance data.

Also, based on study result conducted in this study, It is need to proceed to study about having the effect of performance improvement with changed some factor after constructing IT collaboration system.

Categories and Subject Descriptors

H. [Information Systems]: Miscellaneous

General Terms

Management, Measurement, Performance

Keywords

ICT collaboration, Collaborative network, Automobile business, Analysis performance, the level of collaboration

1. INTRODUCTION

It is considered for companies to make up new types of organization, adopt ICT technologies tailored to the features of company, or etc, as a plan for the company to survive in the competitive environment. Especially, Utilizing IT technologies, when companies do their business, can make companies share their information, and it leads to reduce their costs in the business process and shorten their work time. Namely, introducing and utilizing ICT collaborative system that can make companies collaborate on their business brings about economic collaborative performance for the companies. So, a mother company and its partner companies bring ICT collaborative systems in and share

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their information that is needed by each company to secure their future value, and also try to lead the competitiveness of companies in the maximum level by constructing the collaborative network between a mother company and its partner companies.

In the 21c business environment, the competition between individual companies has being shifted to the competition among Value Chain or Supply Chain. The companies can secure a competitive advantage when they try to improve the outcome of each other through organic collaboration with each other in the competitive environments where uncertainty of supply and demand increases, markets become globalized, product life cycle is shorten, process and product technology are rapidly changed, and so on.

In such circumstances, many companies recognize the importance of collaboration so that the meaning of the relationship with providers is changed from the relationship in the traditional market of business to cooperative relationship.

Collaboration means working together with two or more members of supply chain to create economic benefits through information sharing, collaborative decision making, and shared interest. It is also a process for decisions among organizations that depend on each other, and a concept that includes corporate responsibility for co-owned and results. It is distinguished from "market participants" in respect that organizations form a relationship that is long-term and reliable, not temporary and "hierarchical relationships" in respect that it does not directly control. Specifically it means performing product development, demand forecasting, production planning, distribution management and so on in cooperation with supply company and cooperative companies.

Inter-Organizational Information System signifies automated information systems(e.g. common application program, common database and communication network, etc.) to interconnect business processes among two or more organizations. It also supports Interorganizational collaboration, providing a continuous interchange system where both structured and unstructured data is shared through networks, and promote expansion of business scope and rebuilding interorganizational business process.

Organizations have constructed and utilized IT Collaboration System to create a collaboration network, which consists of materials management, production management, and distribution system in the automobile industry this paper's subject of investigation and analysis. Through this system, organizations share their ordering information, inventory information, operating status of the production line, and so on and apply these shared information to producing products.

In particular, the annual production capacity of Korea's automobile industry is the fifth largest of the world level and the percentage of the total exports is more than 3%. However, it is important for organizations to have competitive power derived from cost reduction as well as superior technology and reliability to survive the competition of the automobile industry due to China's technical capabilities and conclusion of world trade agreements. Therefore, individual companies make an effort to improve the productivity and accuracy by introducing IT collaboration system or ERP(Enterprise Resource Planning) solutions, etc.

Automobile production process(Figure 1) is largely composed of five steps; Pressing process(cutting iron plate and compression

molding) → Body Assembly(welding iron plate, assembly) → Painting(soundproofing, anti-vibration, anti-rust treatment, and paint) → Designing → Final test. Designing&Assembly line is a process where thousands kinds of components conveyed from the painting line are assembled and installed, and completed into the car.



Figure 1. A process of completed car

It is essential for providers to smoothly supply components in the assembly line because the interruption(defective, etc.) in the certain workplace leads to discontinue the work of a full-line. Also, in the automobile industry there are a great variety of components such as materials, half-finished goods, goods-in-process, and product for final product, and the number of partner companies which deliver products to the one of the parent companies is larger than other industries. As for the parent company, it has a high reliance on partner companies. For these reasons, especially in the automobile industry the need for the collaboration system utilizing IT is continuously increasing.

As you look at the production status of the automobile industry group that participated in this research, its own production scale for the production of a finished product automobile was about 35%, and sales scale of partner companies was at least 3 billion or more. In terms of annual collaboration scale, the number of transactions and partners, and the amount of transactions has gradually increased. In such a situation, it is imperative to ensure market-oriented flexible production system, maintain the whole optimal inventory levels of the supply chain between the large, small and medium-sized enterprises, and rationalize procurement and marketing logistics for domestic automobile industry to dominate competitive advantage in the global market.

In terms of collaboration achievement, more than 10% of improvement outcome appears in both the parent company and its partner companies compared to previous systems which didn't have the IT collaborative system.

In terms of the types of collaborative information, we found that the highest degree of performance improvement is applicable to the organizations which share production schedule information.

In the future, it is need to proceed the study of deriving decision factor in the various view for calculating the optimum performance of business group participating(mother company - composition of cooperation company)

For example, there are industrial character, composition of business group, size of cooperation company, sale rate of cooperation company compared to mother company, and orders of cooperation compared to the parent company.

In this study, we collected and analyzed basic data for the target of survey and business manager of the parent company but It is need to proceed analysis of business performance which is active and has validity from the step of business agreement to the step after the end of business rather than submit the data for

performance tracking obligatorily for acquiring an objective performance data.

Also, based on study result conducted in this study, It is need to proceed to study about having the effect of performance improvement with changed some factor after constructing IT collaboration system.

2. RELATED WORK

In Song In Kuk's study, in the environment of information system the improvement of service quality make system and information quality advanced, which results in a positive correlation with not only satisfaction but also outcome of the organization.

The survey items that measure collaborative outcomes are composed of the extent of having the same kind of business mind, of collaboration with each other for solving problems, and of sharing the information. [15]

In Oh Segu's study to measure the organization's performance which arises from the collaboration environment, the authors include the concept of enhancement of national competitiveness, market sales, and satisfaction of partner companies. Specifically,

As a measurement variable for measuring the concept of corporate performance, the authors employ total three survey items as follows: competitiveness has been strengthened after construction of the collaboration environments, satisfaction of trading companies has increased after construction of the collaboration environment, and sales has increased after construction of the collaboration. [13]

Collaborative Process Performance Management Framework(cPPM) includes a process of developing Collaborative Key Performance Indicators(cKPI), monitoring real-time collaborative process, and analyzing collaboration performance.

To constitute the performance management and performance indicator of supply chain SCOR model can be used. This SCOR model provides four performance items as follows: Reliability, Flexibility & Responsiveness, Cost, and Asset, which help the performance of supply chain measured. Also, it presents performance indicators of lower level based on the upper performance indicator hierarchically. These can be measured and analyzed in conjunction with the detailed process of the supply chain. In addition, SCOR model designs supply chain to meet the overall goal of the supply chain and provides a systematic methodology that reflects the measured results. [1]

In Lim Chunseong's study, the authors define the degree to which the user takes advantage of the information and the indicators to assess the level of utilization include the usage of the system, the proportion of the system utilization, the efficiency level of business operations, improvement of job performance and a mind to utilize system. [12]

Jan Recker, JanMendling and ChristopherHahn reported on an empirical study of collaborative process modeling on basis of collaborative modeling technology. They advanced and tested a Capabilities- Gains-Outcomes model, which suggests a set of key technology capabilities that support three stages of the collaborative modeling process, the cognitive process gains this support enables, and its impact on outcomes of the collaboration in terms of quality of the model produced and evaluations of the

technology. They found that the effect of technology capabilities on the quality of models produced is fully mediated by process gains (i.e., enablement and knowledge develop-ment), thus stressing the importance of viewing technol- ogy as an enabler of collaborative work processes. Further, the ease of modeling construct as a technology capability appears to be of central importance for the perceived usefulness of a collaborative modeling tool. Next, we found empowerment and knowledge development to be equally important in predicting the perceived quality of process models. Additionally, empowerment builds on the colla- boration support offered by a tool, as much as knowledge development requires suitable modeling and validation features to be provided. Finally, in terms of outcomes, tool satisfaction and usefulness are independent from the quality of the produced models. Overall, our research draws a picture of collaborative technology as an enabler of process modeling in distributed settings, with effects that are mediated through gains in the collaboration process itself.[7]

Christopher Durugbo analyses integrated information flow in the context of CNs and proposes a conceptual framework to manage flow integration using insights from the case study. Skill, project, agreement and relationship orientations were uncovered as important avenues for managing integrated flow. In view of these orientations, there is a need for managers of CNs to prevent fluidity of collaborative agreements, procedurally generate prompts, create implementation checklists, strive to enhance confidence and relationships with partners, systematise communication templates and electronic mails, formulate strict issuance policies and separate business and technological concerns during collaboration with partners.[6]

R. Mark Koan, Ed.D helped develop a network of IT professionals from schools in a community college system and a research university in the same geographic region into a community of practice (CoP) that spanned the boundaries of their organizations. This collaboration allowed members from participating institutions to share knowledge and ideas relating to shared technical problems. The author also collected data about the extent to which the community developed the factors that contributed to its development and the perceived value that the other leaders and participants attributed to the endeavor.[8]

3. Analysis of the Factors for IT Collaboration

3.1 Methodology of IT collaboration level and performance analysis among organizations

In this paper, the analysis of the performance was carried out to organize the process of utilization and outcomes of IT collaboration system as Figure 1.[4]

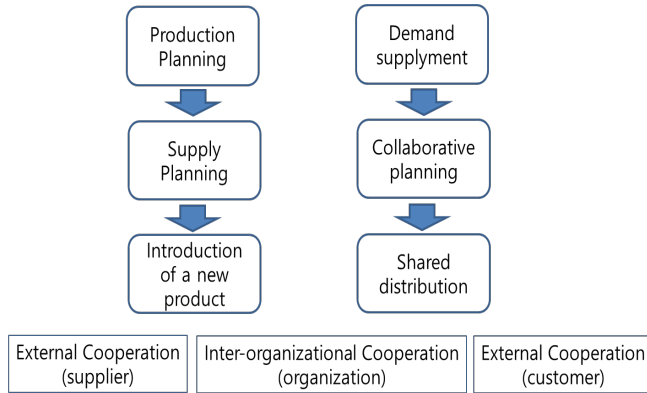


Figure 2. IT Collaboration System Utilization and Performance Process

In addition, we measured operational and financial performance (quantitative results) of order response, Delivery time management, Logistic management system, Inventory management, etc. which can be directly gained from IT collaboration system, and indirect non-financial performance(qualitative results).

Order Response Management is an activity to meet a deadline by planning and controlling the production to fulfill an order(or contract) .

Delivery time management is an activity to comprehend the logistics delivery status of destination and manage loading, quality test, and warehousing not to delay delivery.

Logistic management system is an activity to comprehensively manage the flow of the produced product.

Inventory management is an activity to maintain the inventory assets such as raw materials or product at an appropriate level and supply them to the production line. <Figure 2> [4]

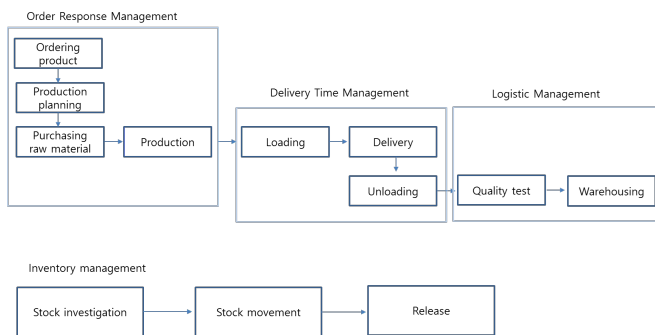


Figure 3. Work Process Scope

3.2 Improvement from IT Collaboration system

The number of parent companies, the research subject of this investigation and analysis, is 3 companies and that of partner companies is 45 companies. The common collaboration system is production and procurement system.

Scale of participating companies in this research (the number of employees) was less than 50people and small-sized enterprises.

Table 1. Investigated Object

The number of parent companies	The number of partner companies	Common collaboration system
3	45	Production, procurement

Table 2. Valid Respondents

Valid Respondents		The number of valid survey/The number of companies participated
The number of parent companies	The number of partner companies	
3	45	48/94

3.2.1 Analysis on sharing cycle of collaborative information

As a result of analysis of sharing cycle of collaborative information through the introduction of the IT collaboration system, the level of the sharing cycle of collaborative information before the parent company and its partner companies used the IT collaboration system was mostly “Not at all” (47.0%), and after they do it the level of it is mostly “partially or frequently share”.(each 33.7%).

Table 3. Analysis on sharing cycle of collaborative information from IT collaboration system

Sharing cycle of collaborative information	Before the introduction of system	After the introduction of system
	ratio(%)	ratio(%)
Not at all	47.0	0.0
Partially share as needed	30.1	8.4
Periodically share	20.5	33.7
Frequently share	2.4	33.7

Real-time share	0.0	24.1
Total	100.0	100.0

3.2.2 Analysis on sharing range of collaborative information

As a result of analysis of sharing range of collaborative information through the introduction of the IT collaboration system, the level of the sharing range of collaborative information after the IT collaboration system is introduced is higher than before it is. Before the introduction of the IT collaboration system, the sharing range of collaborative information was mostly very low. However, after the introduction of the IT collaboration system the sharing range of collaborative information is high or very high.

Table 4. Analysis on sharing range of collaborative information from IT collaboration system

Sharing range of collaborative information	Before the introduction of system	After the introduction of system
	ratio(%)	ratio(%)
Very low	41.7	0.0
Low	20.8	6.3
medium	37.5	25.0
High	0.0	33.3
Very high	0.0	35.4
Total	100.0	100.0

3.3 Performance analysis after using IT collaboration system

Process	Performance Improvement (unit: %)		
	Total	Parent companies	Partner companies
Order Response Management	10.3	18.8	9.9
Delivery Time Management	12.7	10.0	12.8
Logistic Management System	10.0	11.3	9.7
Inventory Management	8.5	11.7	8.3
Total outcome(average)	10.2	10.2	10.2

In this study, the outcome after use of IT collaboration system was measured along business process steps. The extent of improvement in overall performance before use of collaboration system was 10.2%. Specifically, Order Response Management is an activity to meet a deadline by planning and controlling the production to fulfill an order(or contract), and a step where product orders are accepted in real-time and measure the shortened order response time(delivery preparation time). In this study, an improvement, 10.3%, has been investigated in the Order Response Management process.

Delivery Time Management is an activity to comprehend the logistics delivery status of destination and manage loading, quality test, and warehousing not to delay delivery, and a step where shortened delivery delays(loading-quality test) are measured. In this study, an improvement, 12.7%, has been investigated in the Delivery Time Management by adjusting logistics arrival time and sharing delivery information.

Logistic Management System is an activity to comprehensively manage the flow of the produced product, and a step where finally reduced delivery cost (transport costs) is measured with the operational business effects such as a reduction in the number of shipment times, an increase in product shipment unit which leads to the accuracy improvement of logistics planning. In this study, an improvement, 10.0%, has been investigated in the Logistic Management System.

Inventory management is an activity to maintain the inventory assets such as raw materials or product at an appropriate level and supply them to the production line, and a step where an reduction in holding item and inventory storage costs(warehouse, administrative expenses(personnel expenses, equipment), etc.) are measured as the proper inventory holdings amount is reduced in accordance with the introduction and utilization of IT collaboration system between organizations. In this study, an improvement, 8.5%, has been investigated in the Logistic Management System.

4. Conclusion

In this study established inter-organizational IT collaboration systems have moving to diffusion phase along with the development stage (life cycle) of conventional information systems.

The parent company and its partner companies share this following collaborative information: demand forecasting(production planning, payment instructions) information > delivery progress information > resources/inventory information > production schedule information > distribution information

Since the number of IT personnel and scale of partner companies is absolutely insufficient, sharing salient collaborative information that can be fully utilized, that the parent company and its partner companies frequently exchange, is more important and significant than just constructing inter-organizational IT collaboration system multifunctional. In addition, in this paper by using IT collaboration system the quality level of collaboration system between the parent company and its partner companies(system stability, convenience, system response time, etc.), the quality level of information from developed collaborative system (collaborative information timeliness, accuracy), and the level of

sharing cycle of collaborative information and sharing range of collaborative information.

In terms of collaboration achievement, more than 10% of improvement outcome appears in both the parent company and its partner companies compared to previous systems which didn't have the IT collaborative system.

In terms of the types of collaborative information, we found that the highest degree of performance improvement is applicable to the organizations which share production schedule information.

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